Komail Ahmad Khalil

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PROFESSIONAL SUMMARY

Highly motivated Software Engineer and Data Science specialist with robust expertise in Full-Stack Development, Applied Artificial Intelligence, and scalable Machine Learning systems. Proven track record in architecting high-performance web applications and engineering complex computer vision and NLP solutions for real-world industry and healthcare environments. Adept at transforming complex technical requirements into intuitive, reliable, and high-impact digital products.

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, HTML/CSS, SQL
Web Technologies:	React, Next.js, Node.js, Flask, FastAPI, RESTful APIs, MySQL
AI & Machine Learning:	Deep Learning, Federated Learning, Vision-Language Models (VLMs), YOLOv8, RAG pipelines (LangChain), NLP, PyTorch, TensorFlow, OpenCV, Scikit-Learn, ONNX
Tools & Workflows:	Git, Web Deployment, Postman, CI/CD, Agile Methodologies, Pandas, NumPy

PROFESSIONAL EXPERIENCE

Software Engineer
Jun 2026 – Present

Khyber College of Dentistry

- Developing and maintaining robust software solutions to streamline clinical workflows, patient data management, and operational efficiency within a high-volume dental hospital.
- Collaborating with cross-functional healthcare teams to architect and deploy modern, secure, and scalable web applications.
- Ensuring rigorous data security protocols and optimizing system architectures for high availability and low-latency performance in a medical setting.

Full Stack Developer
Nov 2025 – Nov 2026

University of Engineering & Technology (UET) Mardan

- Led the comprehensive redesign and full-stack development of the official university web portal, winning a highly competitive university-wide web design competition.
- Successfully transitioned the platform from a static prototype to a highly responsive, database-driven web application tailored for thousands of active student and faculty users.

AI & Data Science Intern
Dec 2024 – May 2025

AI and Data Science Career Launchpad Program

- Engineered predictive healthcare solutions, including an automated Anemia Detection Platform and a Heart Disease prediction system with an integrated NLP therapist chatbot.
- Executed end-to-end machine learning pipelines using Python, Pandas, and Scikit-Learn, encompassing rigorous data preprocessing, EDA, and model evaluation.
- Collaborated with cross-functional teams to assist in research-oriented tasks, author technical reports, and deliver formal presentations of analytical findings.

RESEARCH & TECHNICAL PROJECTS

Federated Vision-Language Models for 3D Brain Tumor Segmentation 2026

- Applied Deep Learning, Federated Learning (FL), and Vision-Language Models (VLMs) to architect a highly secure, privacy-preserving 3D medical imaging pipeline across heterogeneous clinical institutions.
- Co-authored a highly technical research manuscript evaluating the model's accuracy and robustness, currently under academic peer review.

Real-Time Crime & Incident Detection Using AI Jan 2026 – Jun 2026

- Architected an advanced computer vision system utilizing custom-trained YOLOv8 models to automatically detect severe threats (weapons, fire/smoke) from real-time CCTV video feeds.
- Secured a prestigious provincial Undergraduate Research Support grant from DOST Peshawar, validating the project's technical merit and potential for public safety deployment.
- Authored a research paper detailing the methodology and real-world deployment challenges, currently under review.

AI-Based Anemia Detection Application Apr 2025 – Nov 2026

- Co-developed a comprehensive diagnostic web application integrating clinical test data with RGB pixel value analysis from ocular imagery.
- Engineered a low-latency real-time prediction pipeline utilizing Flask, deploying optimized ONNX models to deliver highly efficient diagnostic inference for end-users.

Multilingual RAG Application & Real Estate Marketplace 2026

- Independently engineered a sophisticated Multilingual Retrieval-Augmented Generation (RAG) system utilizing LLMs, LangChain, and FastAPI to deliver accurate, context-aware AI interactions.
- Developed "Property Hub Pakistan," a full-stack real estate marketplace integrated with MySQL and robust OTP authentication mechanisms.

EDUCATION

Bachelor of Science in Computer Science (Data Science Track) Graduated Jun 2026

University of Engineering & Technology (UET) Mardan

CGPA: 3.20/4.00

FSc / High School in Computer Science Jul 2022

Capital Degree College

Marks: 860 / 1100